Applied Linear Algebra in Data Analysis

Linear Least Squares

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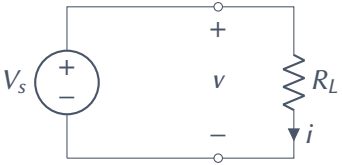
Part I

Overdetermined Systems of Linear Equations

Equations with Tall/Skinny Matrix $\mathbf{A}$

Overdetermined Systems: Motivating Example



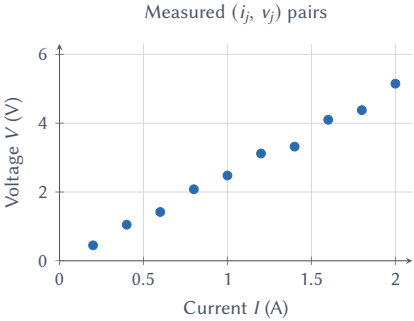
Ohm's law: $v = R_L \cdot i$

Set V_s, read i — repeat n times. Each trial gives one equation:

$$v_j = R_L \cdot i_j, \quad j = 1, \ldots, n$$

n equations, **1** unknown R_L.

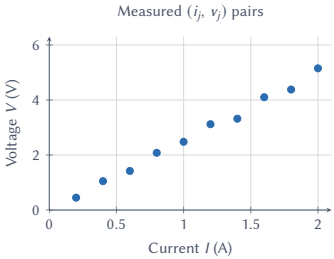
$$\implies \mathbf{i}\left[R_L\right] = \mathbf{v}$$



Points don't lie on any single line $\Rightarrow$ no solution!

Goal: find R_L that best fits the data $\implies$ **find an R_L that minimizes the error in describing the data.**

Overdetermined Systems: Motivating Example



Error in describing the j-th data point:
$e_j = v_j - R_L \cdot i_j$

$$\text{Total Error} = \|\mathbf{e}\|_2^2 = \left\|\mathbf{v} - \mathbf{i}\left[R_L\right]\right\|_2^2$$

where, $\mathbf{e} = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix}^\top$.

Minimizing this total error is the **the linear least squares problem**.: we are minimizing the **sum of squares** of the inidividual errors e_j.
We can express the problem as the following,

$$\text{minimize}_{R_L \in \mathbb{R}} \quad \left\|\mathbf{v} - \mathbf{i}\left[R_L\right]\right\|_2^2$$

Overdetermined System: More than one unknown

For a tall, skinny matrix $\mathbf{A} \in \mathbb{R}^{p \times q}$ ($p > q$), the system $\mathbf{Ax} = \mathbf{b}$ has a solution *only* when $\mathbf{b} \in C(\mathbf{A})$.

- We find an *approximate* solution $\hat{\mathbf{x}}$ when no exact $\mathbf{x}$ exists.
- Define the **least squares problem**: choose $\hat{\mathbf{x}}$ that minimizes

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^q} \|\mathbf{b} - \mathbf{Ax}\|_2^2$$

- $\mathbf{A}$ and $\mathbf{b}$ come from data. The quantity $\|\mathbf{b} - \mathbf{Ax}\|_2^2$ is called the **objective function**.

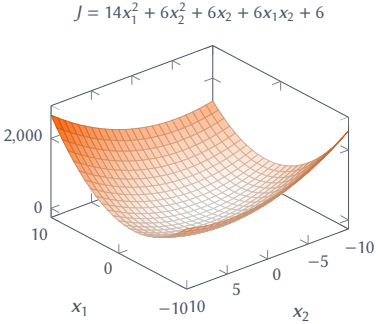
When $\mathbf{b} \in C(\mathbf{A})$ the minimum is zero and we recover an exact solution.

Least Squares Problem

Two-dimensional example

$$\left.\begin{aligned} 2x_1 - x_2 &= 2 \\ -x_1 + x_2 &= 1 \\ 3x_1 + 2x_2 &= -1 \end{aligned}\right\} \longrightarrow \mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \\ 3 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$\|\mathbf{b} - \mathbf{Ax}\|^2 = 14x_1^2 + 6x_2^2 + 6x_1x_2 + 6 \geq 0$$



The objective function attains its minimum at

$$\hat{x}_1 = \frac{52}{75}, \quad \hat{x}_2 = \frac{3}{25}$$

Part II

Normal Equations and Geometric Interpretation

Least Squares: Normal Equations

The general solution is derived by setting the gradient to zero. Let $f(\mathbf{x}) = \|\mathbf{b} - \mathbf{A}\mathbf{x}\|_2^2$.

$$f(\mathbf{x}) = \mathbf{b}^\top \mathbf{b} - 2\mathbf{b}^\top \mathbf{A}\mathbf{x} + \mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x}$$

At the minimum, the gradient $\nabla f(\mathbf{x})^\top$ is zero:

$$\nabla f(\mathbf{x})^\top = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix} = \mathbf{0}^\top$$

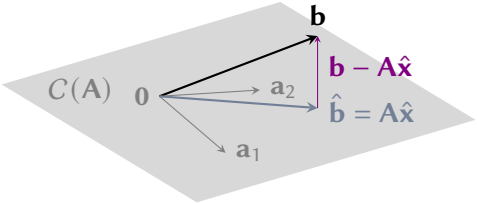
Working through the algebra yields the **Normal Equations**:

$$\nabla f(\mathbf{x})^\top = -2\mathbf{b}^\top \mathbf{A} + 2\mathbf{x}^\top \mathbf{A}^\top \mathbf{A} = \mathbf{0}^\top \implies \underbrace{\mathbf{A}^T \mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}}_{\text{Normal Equations}}$$

When $\mathbf{A}$ is **full rank**, we have a closed-form solution:

$$\hat{\mathbf{x}} = \left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}^\dagger \mathbf{b}$$

Least Squares: Geometric Interpretation



Two key facts about $\hat{\mathbf{b}} = \mathbf{A}\hat{\mathbf{x}}$:

1. $\hat{\mathbf{b}} \in C(\mathbf{A})$
2. $\|\mathbf{b} - \hat{\mathbf{b}}\|_2^2$ is minimized

Minimality implies $(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) \perp \mathbf{A}\hat{\mathbf{x}}$:

$$\hat{\mathbf{x}}^\top \underbrace{\left(\mathbf{A}^\top \mathbf{b} - \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}}\right)}_{\text{Normal Equations}} = 0$$

*The least squares solution $\hat{\mathbf{x}}$ solves $\mathbf{A}\mathbf{x} = \hat{\mathbf{b}}$, where $\hat{\mathbf{b}}$ is the orthogonal **projection** of $\mathbf{b}$ onto $C(\mathbf{A})$.*

Part III

Multiobjective Linear Least Squares

When we have more than one criteria to optimize

Multi-Objective Least Squares

Motivation

Many applications involve **multiple, potentially conflicting objectives**:

$$J_1 = \|\mathbf{A}_1\mathbf{x} - \mathbf{b}_1\|^2, \quad J_2 = \|\mathbf{A}_2\mathbf{x} - \mathbf{b}_2\|^2, \quad \ldots \quad J_k = \|\mathbf{A}_k\mathbf{x} - \mathbf{b}_k\|^2$$

- Combine into a single weighted objective:

$$J = \sum_{i=1}^{k} \rho_i J_i, \quad \rho_i > 0 \ \ \forall\, i$$

- The weights ρ_i encode the **relative importance** of each objective.
- Normalizing by J_1:

$$J = J_1 + \sum_{i=2}^{k} \rho_i J_i$$

Multi-Objective Least Squares

Reduction to a single least squares problem

$$J = \rho_1 \left\|\mathbf{A}_1\mathbf{x} - \mathbf{b}_1\right\|^2 + \cdots + \rho_k \left\|\mathbf{A}_k\mathbf{x} - \mathbf{b}_k\right\|^2 = \left\|\sqrt{\rho_1}\mathbf{A}_1\mathbf{x} - \sqrt{\rho_1}\mathbf{b}_1\right\|^2 + \cdots + \left\|\sqrt{\rho_k}\mathbf{A}_k\mathbf{x} - \sqrt{\rho_k}\mathbf{b}_k\right\|^2$$

$$J = \left\| \begin{bmatrix} \sqrt{\rho_1}\mathbf{A}_1 \\ \sqrt{\rho_2}\mathbf{A}_2 \\ \vdots \\ \sqrt{\rho_k}\mathbf{A}_k \end{bmatrix} \mathbf{x} - \begin{bmatrix} \sqrt{\rho_1}\mathbf{b}_1 \\ \sqrt{\rho_2}\mathbf{b}_2 \\ \vdots \\ \sqrt{\rho_k}\mathbf{b}_k \end{bmatrix} \right\|^2 = \left\|\tilde{\mathbf{A}}\mathbf{x} - \tilde{\mathbf{b}}\right\|^2 \implies \hat{\mathbf{x}} = \left(\tilde{\mathbf{A}}^\top \tilde{\mathbf{A}}\right)^{-1} \tilde{\mathbf{A}}^\top \tilde{\mathbf{b}}$$

The columns of $\tilde{\mathbf{A}}$ are independent if the columns of *at least one* $\mathbf{A}_i$ are independent.

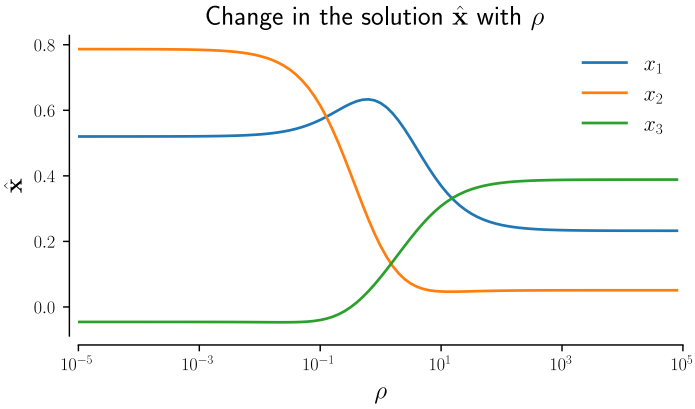
For the two-objective case $J = J_1 + \rho J_2$:

$$\hat{\mathbf{x}} = \begin{cases} \operatorname{argmin}_{\mathbf{x}} \left\|\mathbf{A}_1\mathbf{x} - \mathbf{b}_1\right\|^2 & \rho = 0 \\ \operatorname{argmin}_{\mathbf{x}} \left\|\mathbf{A}_2\mathbf{x} - \mathbf{b}_2\right\|^2 & \rho \to \infty \end{cases}$$

Multi-Objective Least Squares

Solutions as ρ varies

Consider $J(\mathbf{x}) = J_1(\mathbf{x}) + \rho\, J_2(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^3$.



Each value of ρ selects a different trade-off between the two objectives.

Multi-Objective Least Squares

Pareto optimality

Any solution on the **Pareto-optimal curve** is optimal in the sense that there exists no $\tilde{\mathbf{x}}$ such that

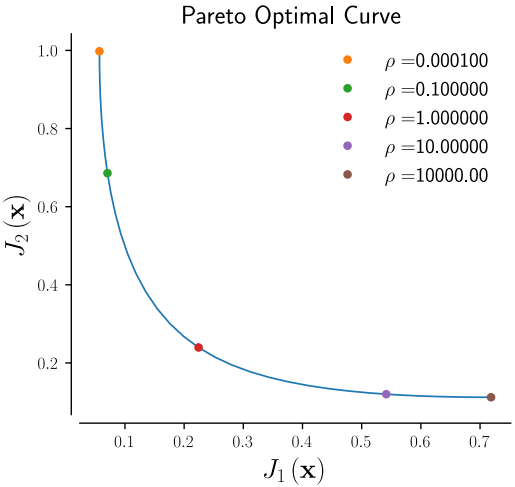
$$J_1(\tilde{\mathbf{x}}) \leq J_1(\hat{\mathbf{x}})$$

and

$$J_2(\tilde{\mathbf{x}}) \leq J_2(\hat{\mathbf{x}})$$

with both inequalities strict.

Improving one objective necessarily worsens the other.



Regularized Least Squares

Multi-objective LS is also useful when individual $\mathbf{A}_i$ lack full column rank.

- Suppose $\mathbf{A}_1 \in \mathbb{R}^{m_1 \times n}$ and $\mathbf{A}_2 \in \mathbb{R}^{m_2 \times n}$ with $m_1,\ m_2 < n$ (neither has independent columns). If $m_1 + m_2 \geq n$, the stacked matrix $\tilde{\mathbf{A}}$ *can* have independent columns.
- **Tikhonov regularization**: add a penalty on the norm of $\mathbf{x}$,

$$J = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \rho \|\mathbf{x}\|^2, \quad \rho > 0$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \\ \sqrt{\rho}\,\mathbf{I} \end{bmatrix} \implies \hat{\mathbf{x}} = \left(\mathbf{A}^\top \mathbf{A} + \rho\,\mathbf{I}\right)^{-1} \mathbf{A}^\top \mathbf{b}$$

Tikhonov Regularization

$$J = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \rho\,\|\mathbf{x}\|^2, \quad \rho > 0 \qquad \hat{\mathbf{x}} = \left(\mathbf{A}^\top \mathbf{A} + \rho\,\mathbf{I}\right)^{-1} \mathbf{A}^T \mathbf{b}$$

- $\hat{\mathbf{x}}$ is unique even when $\mathbf{A}$ is *not* full rank, because $\mathbf{A}^T\mathbf{A} + \rho\,\mathbf{I}$ is always invertible for $\rho > 0$.
- Even when $\mathbf{A}$ is full rank, the regularization term **improves the condition number** of the matrix, yielding numerically more stable solutions.

Regularization trades a small increase in residual for a large improvement in numerical stability and uniqueness.

Constrained Least Squares

Problem: minimize the residual subject to equality constraints,

$$\begin{aligned} \text{minimize} \quad & \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 \\ \text{subject to} \quad & \mathbf{C}\mathbf{x} = \mathbf{d} \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{p \times q}$, $\mathbf{C} \in \mathbb{R}^{l \times q}$, $\mathbf{b} \in \mathbb{R}^p$, $\mathbf{d} \in \mathbb{R}^l$.

Solved via **Lagrange multipliers**, leading to the **KKT (Karush–Kuhn–Tucker) equations**:

$$\begin{bmatrix} 2\left(\mathbf{A}^T\mathbf{A}\right) & \mathbf{C}^T \\ \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ \hat{\mathbf{z}} \end{bmatrix} = \begin{bmatrix} 2\mathbf{A}^T\mathbf{b} \\ \mathbf{d} \end{bmatrix}$$

The $(q + l) \times (q + l)$ coefficient matrix is invertible if and only if $\begin{bmatrix} \mathbf{A} \\ \mathbf{C} \end{bmatrix}$ is full rank.